

Operating Systems 2025 (2025/04/30)

Please answer questions in order and leave blank for unanswered questions

There are 17 questions

1. (6%) Please give 3 basic approaches for Multiprocessor (MP) operating systems.

Ans.

- (1). Each CPU Has Its Own Operating System.
- (2). Master-Slave Multiprocessors (MS-MP).
- (3). Symmetric Multiprocessors (SMP).

2. (8%) What are the major pros and cons of non-blocking primitives in message passing systems?

Ans.

Pros: (4%)

- (1). The sender and/or the receiver can continue their executions without waiting for each other.
- (2). The system is less likely to run into a deadlock.

Cons: (4%)

- (1). The sender does not know whether or not the receiver has actually received the message.
- (2). The sender cannot modify the message buffer until the message has been sent.

Additional overhead is required to notify the sender and the receiver.

Additional data copy is required to reuse message buffer.

3. (6%) For crossbar networks and omega networks (also called perfect shuffle), which network may occur blocking? Why do we still need the network that may occur blocking?

Ans.

a. (2%) Omega networks may occur blocking, but we still need it because of its cost.

b. (4%) Crossbar network needs n^2 crosspoints, but Omega network only needs $\frac{n}{2} \log_2 n$ switches.

4. (4%) A distributed system is described as scalable when it remains effective when there is a significant increase in the number of resources and the number of users. However, these systems sometimes face performance bottlenecks. How can these be avoided?

Ans.

Caching and replication may be used to improve the performance of the resources that are very heavily used in a distributed system.

5. (8%) Please list and describe 4 consistency semantics for file sharing a distributed system.

Ans. (2% each)

1. Sequential consistency (Lamport, 1979):

“The result of any execution is the same as if the operations of all the processes were executed in some sequential order and the operations of each individual process appear in this sequence in the order specified by its program”

2. Causal consistency (Hutto and Ahamad, 1990):

“Writes that are potentially causally related must be seen by all processes in the same order. Concurrent writes may be seen in a different order on different machines”

3. FIFO consistency:

“Writes done by a single process are seen by all other processes in the order in which they were issued, but writes from different processes may be seen in a different order by different processes” (PRAM consistency, Lipton and Sandberg 1988)

4. Session consistency was proposed and used in AFS.

The AFS server keeps a list of which client has which file.

All updates are written back to the server on file close.

On close, if a file has been changed, the server notifies all clients with a copy.

The client can decide either reopen it to get the latest version or continue with the old version.

Therefore, updates are visible only after file close.

On opening a file, a client gets the last closed version.

6. (4%) Please List two disadvantages of the Direct Communication Model.

Ans.

(1). Changing the name of a process requires all related processes to update their programs.

(2). Process names need to be known in advance.

7. (8%) In multicasting systems, there are 4 level of reliability degrees that can be specified by a sender. Please list and describe them briefly.

Ans. (2% each)

(1). The 0-reliable. No response is expected from any of the receivers.

An example of this type of application is a time signal generator.

(2). The 1-reliable. Only one response is expected from any of the receivers.

Sends a request to several agents and selects the fastest for the future request.

(3). The M-out-of-N-reliable.

The multicast group consists of N receivers and the sender expects M responses from receivers, where $1 < M < N$.

Erasure coding, RAID, etc.

(4). All-reliable. The sender expects a response from all the receivers.

8. (4%) What is the cache coherence problem?

Ans.

When a CPU wants to write a word which is in more than one caches, the system must inform all caches of the write request.

if a cache has a clean copy, it can just discard the copy.

If a cache has a "dirty" (modified) copy, the cache must either write the modified copy back to the memory before the write request can proceed or transfer the modified copy to the writer directly.

This set of rules for maintaining the cache consistency is called cache coherence protocol.

9. (8%) Suppose that the switch 2B and 3C in the omega network of Fig. 8-5 fail. What CPUs are cut off from which memories? Please list all pairs of (CPU, MEM) that cannot communicate.

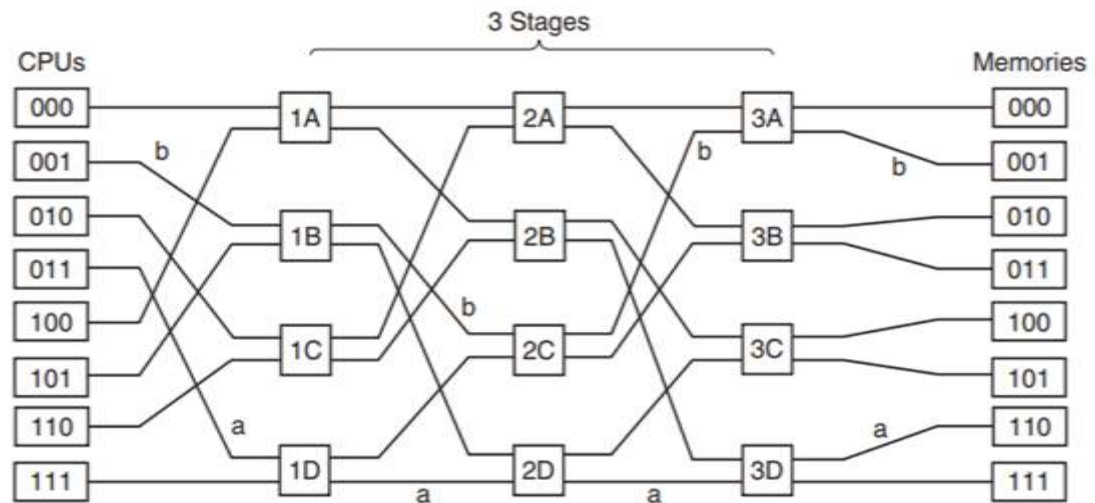


Figure 8-5. An omega switching network.

Ans. (1% for each CPU)

(000, 100), (000, 101), (000, 110), (000, 111),
 (001, 100), (001, 101),
 (010, 100), (010, 101), (010, 110), (010, 111),
 (011, 100), (011, 101),

(100, 100), (100, 101), (100, 110), (100, 111),
 (101, 100), (101, 101),
 (110, 100), (110, 101), (110, 110), (110, 111)
 (111, 100), (111, 101),

10. (6%) Use the omega network in the above, please give the routes for the following CPU memory pairs. (hint: CPU 000 → 1A → 2A → 3A → Memory 000)

- (a) from CPU 010 to Memory 101
 (b) from CPU 100 to Memory 111
 (c) from CPU 111 to Memory 010

Ans. (2% each)

(a) 010 → 1C → 2B → 3C → 101

(b) 100 → 1A → 2B → 3D → 111

(c) 111 → 1D → 2C → 3B → 010

11. (6%) For each of the topologies of Fig. 8-16, what is the bisection width of the interconnection network?

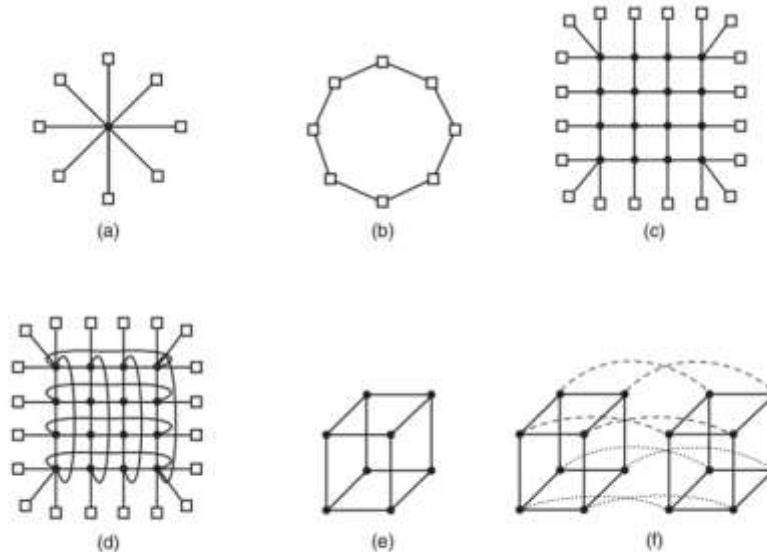


Figure 8-16. Various interconnect topologies. (a) A single switch. (b) A ring. (c) A grid. (d) A double torus. (e) A cube. (f) A 4D hypercube.

Ans. (1% each)

(a) $O(N)$ or 4 (b) 2 (c) $O(\sqrt{N})$ or 4 (d) 8 (e) $O(N)$ or 4 (f) 8

12. (4%) What's the difference between Open and Closed Groups?

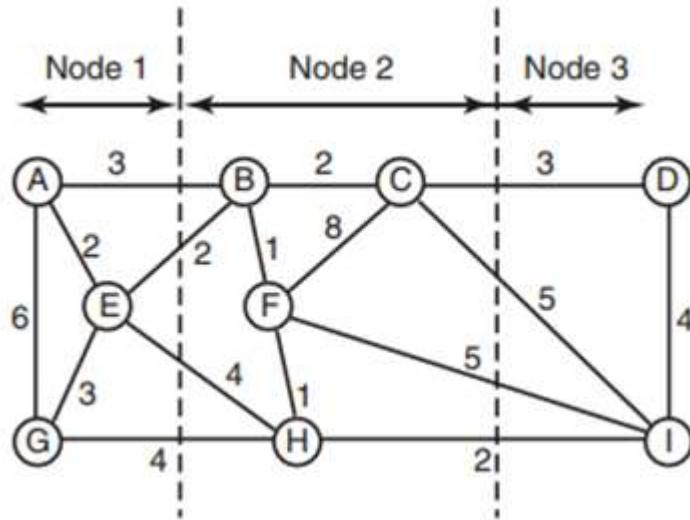
Ans.

(1). A closed group is one in which only the members of the group can send a message to the whole group. An outside process cannot send a message to the

group as a whole, although it may send a message to an individual member of the group.

(2). An open group is one in which any process in the system can send a message to the whole group.

13. (5%) Consider the processor allocation of the following figure. What is the total weight of the external traffic? If process F is moved from node 2 to node 3. What is the total weight of the external traffic now?



Ans.

- a. (2%) the cut between node 1 and 2 contains AB, BE, EH, and GH for a weight of 13. The cut between node 2 and node 3 contains CD, CI, FI, and HI for a weight of 15. The cut between node 1 and 3 contains no edges. The total weight is 28.
- b. (3%) In this split, node 1 has A, E, and G, node 2 has B, C, and H, and node 3 has F, D, and I. The cut between nodes 1 and 2 now contains AB, BE, EH and GH for a weight of 13. The cut between nodes 2 and 3 now contains BF, CF, FH, CD, CI, and HI for a weight of 20. The cut between nodes 1 and 3 now contains no edges. The total weight is 33.

14. (6%) Describe how to implement a binary semaphore by using the Linda system (You can use the following command: out, in).

Ans. (2% each)

Initialization: out("sem")

wait(): in("sem")

signal(): out("sem")

15. (8%) Please list 4 common failures may happen in RPC model?

Ans.

- (1). Cannot locate the server
- server down
- version mismatch
- other exceptions
- (2). Request message is lost
- (3). Reply message is lost
- (4). Server crashes after receiving a request
- (5). Client crashes after sending a request

16. (6%) Please list 3 possible problems for passing parameters in RPC calls.

Ans. (2% each)

- (1). passing pointer: copy on demand (useless for complex structures)
- (2). passing unbounded array: size of an array in C can be undefined in compiling time
- (3). unbounded number of parameters: printf in C has different number of parameters
- (4). using global variables as parameters.

17. (6%) Please list 3 common approaches for identifying a process in direct communication based message passing systems.

Ans. (2% each)

- (1). "Hostname + Local-ID "
- (2). " Hostname (original) + Local-ID + Hostname (current) "

Initially, the 2nd hostname and the 1st hostname are both the hostname of the original creator.

When a process migrates to a new host, the 2nd hostname is changed to the hostname of the new host.

Every host maintains a mapping table for migration information.

A message is always sent to 2nd host with "1st hostname + local-ID".

Upon receiving a message, if the destination process was moved, the new 2nd hostname is used to forward the message.

The new global process ID can be sent back to the sender on ACK.

- (3). A global name server is used for mapping a process's human readable string name to its current global ID